

JEE MAIN REVISION MASTER: SIMPLE HARMONIC MOTION 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Simple Harmonic Motion - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Simple Harmonic Motion - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Simple Harmonic Motion.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A mass of 1 kg is executing SHM. Its displacement is given by $x=6.0\cos(100t+\pi/4)$ cm. What is the maximum kinetic energy?..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The variations of kinetic energy $K(x)$, potential energy $U(x)$ and total energy as a function of displacement of a particle in SHM is as ..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "For a particle executing simple harmonic motion (SHM), at its mean position..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A mass of 1 kg is executing SHM. Its displacement is given by $x = 6.0 \cos(100t + \pi/4)$ cm. What is the maximum kinetic energy?

- (A) 3 J
- (B) 6 J
- (C) 9 J
- (D) 18 J

Correct Answer: None

Solution Analysis: For SHM, $x = A \cos(\omega t + \phi)$ Given displacement is $x = 6.0 \cos(100t + \pi/4)$ cm So the amplitude is $A = 6.0 \text{ cm} = 0.06 \text{ m}$ and angular frequency is $\omega = 100 \text{ rad s}^{-1}$ In SHM, maximum kinetic energy is equal to the total mechanical energy: $K_{\text{(max)}} = (1/2)m\omega^2 A^2$ Substitute the values: $K_{\text{(max)}} = (1/2)(1)(100)^2(0.06)^2$ $K_{\text{(max)}} = (1/2) \times 10000 \times 0.0036$ $K_{\text{(max)}} = 18 \text{ J}$ So the maximum kinetic energy is 18J Correct option: Option D.